

ITS203 Individual Assignment Idea Proposal: Rupak Balami

Problem Description:

Many libraries still use manual methods such as spreadsheets and paper records to manage their books, members and loan tracking. This method is very time consuming and susceptible to human error. Also, it can be difficult to monitor who has current loans, what books are out and who has overdue books. With the help of software, these tasks would be faster, more accurate and less complicated to execute.

Proposed Idea:

I proposed using C# to build a desktop application with Windows Forms that can be used to control all the critical functions of the library digitally. The programme will allow the staff of the library to register patrons, add books to the library, search for books, manage book borrowing and returns, and view past due books. The system would make use of concepts of Object Oriented Programming like inheritance, polymorphism, encapsulation, abstraction, exception handling etc.

Inspiration:

I picked this project because many individuals use libraries on a daily basis, and it's critical to manage them well. I have been in libraries, and have witnessed the positive impact a well designed system can have. In addition, I believed that this project would give me an excellent opportunity to be able to apply all the OOP concepts I had just learned in course in a meaningful manner.

Functional Requirements:

- Adding, viewing, searching and removing books from the library catalogue.
- Keep up to date member records and enrol new members.
- Automatically process book borrowing and set due dates.
- Process book returns and make books available.
- Display all loans, both current and outstanding.
- Display a dashboard summary report for the library.

Non-Functional Requirements:

- It should be user friendly and easy to use for library staff.
- The programme needs to react to user input fast.
- Error messages should be informative and explanatory.
- Any invalid input should be treated gracefully, not causing the system to crash.
- During regular use, the application has to be dependable and stable.

Database Backend Plan:

For this project, I wanted to use JSON file-based data persistence, so that the data can be stored between sessions without needing to install a separate database. The current implementation, however, saves data in memory for the application session due to time constraints during the development. The data in the sample is preloaded when the application starts. Some additional improvements of this system could be adding JSON or SQLite database storage for the permanent data.